

# Modeling Aggregate Demand for Products and Services based on Consumer Preferences for Product Features

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## Abstract

This article presents a model of aggregate demand for products and services based on consumer preferences for product features. Organizations offering services and products can use this model to prioritize feature development and service improvement initiatives by estimating the potential impact of product changes on aggregate demand.

## Introduction

In this article, we present a mathematical model of aggregate demand for products and services based on consumer preferences for product features. We show how to use this model to estimate the potential impact that changes to product features will have on consumer demand. Using these techniques, organizations can prioritize product development and service improvement efforts.

## The Model

We start by assuming that there is a set of products  $A_i$  and potential consumers in a marketplace. We characterize the products using a set of common feature dimensions, such as weight, cost, and performance. Thus, every product  $i$  is associated with a vector listing its feature  $j$  values  $[x_{j,i}]$ . Every potential consumer  $l$  assigns a different weight to each feature dimension  $i$ . These weights are represented by the consumer's preference vector  $[k_{l,j}]$ . Consumers assign a quality score to each product  $s_{l,i} = \sum_{j=0}^n k_{l,j}x_{j,i}$  that is a linear combination of the product's feature values and the consumer's feature weights.

We assume that, for every feature dimension  $j$ , consumer preference weights  $k_{l,j}$  are normally distributed and that preferences across dimensions are independent.<sup>1</sup>

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<sup>1</sup>In general, preferences will not be independent across feature dimensions. For example, the importance that a consumer assigns to the weight of a product might correlate with the

Because the quality scores are linear combinations of independent normally distributed variables, the quality scores for each product will also be normally distributed.

Let  $\mu_j$  and  $\sigma_j$  represent the mean and standard deviation of weights that consumers give to feature dimension  $j$ . The distribution of consumer quality scores  $s_i$  for a product  $A_i$  will equal <sup>23</sup>

$$s_i \sim \mathcal{N}(\sum_{j=0}^n x_{j,i} \mu_j, \sum_{j=0}^n x_{j,i}^2 \sigma_j^2). \quad (1)$$

When presented with two products  $A_i$  and  $A_j$ , a potential consumer will prefer  $A_i$  over  $A_j$  when  $s_i > s_j$ . If  $s_i = s_j$ , the consumer will be indifferent between  $A_i$  and  $A_j$  and will choose one of them randomly with probability  $\frac{1}{2}$ .

Let the quality scores  $s_i$  for product  $A_i$  be normally distributed with mean  $\mu_i$  and standard deviation  $\sigma_i$ . Additionally, let the quality scores  $s_j$  for another product  $A_j$  be normally distributed with mean  $\mu_j$  and standard deviation  $\sigma_j$ . We can calculate the probability  $p_{i,j}$  that a consumer will choose  $A_i$  over  $A_j$  as

$$p_{i,j} = \int \int_0^\infty \varphi\left(\frac{s - \mu_j}{\sigma_j}\right) \varphi\left(\frac{s + \delta - \mu_i}{\sigma_i}\right) d\delta ds. \quad (2)$$

(2) is equivalent to

$$p_{i,j} = \Phi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right). \quad (3)$$

The derivation is presented in Appendix 1.

Let  $p$  represent the probability that a randomly selected consumer chooses either product  $A_i$  or  $A_j$ . Then, if there are  $n$  consumers, in  $np$  times, a consumer will choose either  $A_i$  or  $A_j$ . Let  $n_i$  and  $n_j$  represent the number of times that a consumer chooses  $A_i$  and  $A_j$  respectively.

$$\begin{aligned} np \ p_{i,j} &= n_i \\ np \ p_{j,i} &= n_j. \end{aligned}$$

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weight that they assign to its size. Thus, in practice, we will typically need to perform factor analysis such as Principal Component Analysis (PCA) to identify a set of independent feature dimensions.

<sup>2</sup>Note that the second argument to  $\mathcal{N}$  gives the distribution's variance.

<sup>3</sup>Soch, J. (2021, June 2). Proof: Linear combination of independent normal random variables. The Book of Statistical Proofs. <https://statproofbook.github.io/P/norm-lincomb.html>

Equating both equations gives:

$$np = \frac{n_i}{p_{i,j}} = \frac{n_j}{p_{j,i}}$$

Rearranging gives the relative frequency  $k_{i,j}$  with which consumers choose  $A_i$  over  $A_j$ .<sup>4</sup>

$$k_{i,j} = \frac{p_{i,j}}{p_{j,i}} = \frac{n_i}{n_j}$$

We can expand this equation to rewrite  $k_{i,j}$  as follows:

$$k_{i,j} = \frac{\Phi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right)}{\Phi\left(\frac{\mu_j - \mu_i}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right)} = \left(\Phi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right)^{-1} - 1\right)^{-1}.$$

Let  $p_i$  represent the absolute probability that a consumer will choose product  $A_i$ . Then, assuming that every consumer chooses a product, we can form the equation:

$$p_0 + p_1 + \cdots + p_n = 1.$$

We can substitute the likelihood ratios into this equation to form:

$$\begin{aligned} p_0 + k_{1,0}p_0 + \cdots + k_{n,0}p_0 &= 1 \\ \cdots & \\ k_{0,n}p_n + k_{1,n}p_n + \cdots + p_n &= 1 \end{aligned}$$

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<sup>4</sup>We can derive a useful approximation for  $k_{i,j}$ . Let  $x_{i,j} := \frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}$  and observe that  $x_{i,j} = -x_{j,i}$ . Then we can write  $k_{i,j}$  as  $\frac{\Phi(x_{i,j})}{\Phi(-x_{i,j})}$ . When  $|x_{i,j}| \leq 1$ , we can use the third-order Taylor expansion to practically approximate  $k_{i,j}$  as:

$$k_{i,j} \approx -\frac{x_{i,j}^3 - 6x_{i,j} - 3\sqrt{2\pi}}{x_{i,j}^3 - 6x_{i,j} + 3\sqrt{2\pi}} \quad \text{when } |x_{i,j}| \leq 1.$$

Within this domain, the error is generally less than 7%.

We can easily solve these equations to calculate the absolute probability that a consumer will choose a each product. For example:

$$p_0 = \frac{1}{1 + k_{1,0} + \dots + k_{n,0}}.$$

Generalizing we derive the following equation, which returns the absolute probability  $p_i$  that a consumer will choose product  $A_i$  when the mean utility value assigned to  $A_i$  equals  $\mu_i$  and these values have standard deviation  $\sigma_i$ :

$$p_i = \left[ \sum_{j=0}^{n-1} \left( \Phi \left( \frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}} \right) - 1 \right)^{-1} \right]^{-1}. \quad (4)$$

Thus, we can calculate the proportion of consumers who will choose each product offered within a marketplace based on their preferences for product features. Using these equations, we can calculate the effect that feature changes will have on aggregate demand for various products.

### Example Use

Imagine that we want to model the aggregate demand for three computer processors. Each processor is characterized by myriad features however, we determine that four factors, price, power consumption, area, and performance, explain most consumer preference. We conduct a series of surveys and determine that the weights assigned to these factors are independent and normally distributed.

| $j$ | Factor      | Mean $\mu$ | Standard Deviation $\sigma$ |
|-----|-------------|------------|-----------------------------|
| 0   | Price       | -3         | 1                           |
| 1   | Power       | -2         | 3                           |
| 2   | Performance | 4          | 3                           |
| 3   | Area        | -1         | 2                           |

We then measure three computer processors along each of these four dimensions and produce the following measurements.

| Product | Price $x_{0,i}$ | Power $x_{1,i}$ | Performance $x_{2,i}$ | Area $x_{3,i}$ |
|---------|-----------------|-----------------|-----------------------|----------------|
| $A_0$   | 1               | 1               | 1                     | 1              |
| $A_1$   | 10              | 3               | 5                     | 3              |
| $A_2$   | 2               | 2               | 3                     | 2              |

Based on these factor values, we can use (1) to calculate the score distribution for each product.

| Product | Mean Score $\mu$ | Score Standard Deviation $\sigma$ |
|---------|------------------|-----------------------------------|
| $A_0$   | -2               | 4.7958                            |
| $A_1$   | -19              | 21.0238                           |
| $A_2$   | 0                | 11.7047                           |

Once we have derived the distributions for the product quality scores, we can calculate the probability  $p_{i,j}$  that a randomly selected consumer will choose one product over another for each pair of products  $A_i$  and  $A_j$  using (3). We can record these probabilities in a matrix like the following

$$P_{i,j} := \Phi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right)$$

$$P = \begin{bmatrix} 0.5000 & 0.7848 & 0.4372 \\ 0.2152 & 0.5000 & 0.2149 \\ 0.5628 & 0.7851 & 0.5000 \end{bmatrix}.$$

Dividing along the diagonals, we can calculate the likelihood ratios  $k_{i,j}$  for each pair of products

$$K_{i,j} := \left(\Phi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right)^{-1} - 1\right)^{-1}$$

$$K = \begin{bmatrix} 1.0000 & 3.6459 & 0.7768 \\ 0.2743 & 1.0000 & 0.2737 \\ 1.2874 & 3.6538 & 1.0000 \end{bmatrix}.$$

Adding elements within each column, we can calculate the probability that a randomly selected consumer will choose each product

$$A_i := \left[ \sum_{j=0}^{n-1} \left( \Phi \left( \frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}} \right) - 1 \right)^{-1} \right]^{-1}$$

$$A_0 = \frac{1}{1.0000 + 0.2743 + 1.2874} = 0.3904$$

$$A_1 = \frac{1}{3.6459 + 1.0000 + 3.6538} = 0.1205$$

$$A_2 = \frac{1}{0.7768 + 0.2737 + 1.0000} = 0.4877$$

However, we can do more than calculate relative market share for each product. We can go further and calculate the impact that changes to product features will have on market share. For example, we can use this model to calculate the price that maximizes the gross earnings for the company that produces the second product  $A_1$ .

Let's assume that the market size consists of 10 million units and that the following function is used to convert price scores into actual prices

$$price = 5(e^{priceScore/10} - 1)$$

Then each product has the following gross earnings

| Product | Gross Earnings  |
|---------|-----------------|
| $A_0$   | \$2,052,795.88  |
| $A_1$   | \$10,351,453.05 |
| $A_2$   | \$5,398,845.97  |

However, our model indicates that the company that manufactures  $A_1$  will actually maximize their gross earnings if they can reduce their price from \$8.59 to \$6.56. Doing so will increase their market share to 16.06% and increase their gross earnings to \$10,529,874.61.

## Related Work

We have derived a choice model based on the following assumptions:

1. consumers use linear models to assess the value/utility of products
2. the factors within these models are independent

3. the weights that consumers assign to these factors are normally distributed.

The seminal model of consumer choice is that proposed by the Nobel Laureate Daniel McFadden. McFadden's 1974 article entitled "Conditional Logit Analysis of Qualitative Choice Behavior" assumed that consumer utility values included error terms that were distributed according to the Gumbel distribution.<sup>5</sup>

The model presented here is similar to Multinomial Probit models in that it assumes that the pairwise preference probability is given by the normal cumulative density function. However, we do not conceptually have error terms in our model.

## Conclusion

In this article, we have presented a mathematical model of aggregate product demand based on consumer preferences for product features. We have shown how to use this model to calculate the effect that product changes will have on consumer demand. This knowledge can be used to prioritize product development initiatives and to support planning.

While we have focused on products, the same methods can be used to model demand for services. Additionally, while our examples have assumed that the total number of units sold within a market is fixed, we can easily extend the model to represent variable demand by introducing a "null" product.

In practice, the primary difficulty using this model, as is often the case, lies in parameterization. Unfortunately, it is not possible to "run the equations backwards" and infer the input variables such as product feature scores from output variables such as market share. In general, there are many different product "configurations" that will produce the same market share divisions. As a result, the methods introduced in this article are best used for qualitative and approximate modeling. Input variables such as product scores, and consumer weights, can be approximated using consumer surveys and expert judgement.

Lastly, we hope that our mathematical solutions to the equations are valuable. While we are confident that these equations have been presented and solved in other contexts, we were unable to find solutions to them in our readings. Hence, we hope that this article will make their derivation and proofs easier to find.

## Appendix 1: Solving the Choice Equation

Assume that a population of consumers choose between two products  $A_i$  and  $A_j$ . Every consumer assigns a quality score  $s_i$  and  $s_j$  to products  $A_i$  and  $A_j$  respectively. These quality scores are normally distributed with means  $\mu_i$  and  $\mu_j$  and standard deviations  $\sigma_i$  and  $\sigma_j$  respectively. A consumer will choose  $A_i$  over  $A_j$  if  $s_i > s_j$ . If  $s_i = s_j$  the consumer will be indifferent and will choose one

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<sup>5</sup>McFadden, Daniel F. (1974). "Conditional Logit Analysis of Qualitative Choice Behavior" (PDF). Archived (PDF) from the original on 2000-09-30. Retrieved October 25, 2019.

of them randomly with probability  $\frac{1}{2}$ . The probability that a randomly selected consumer will choose  $A_i$  over  $A_j$  is given by

$$p_{i,j} = \int \int_0^\infty \varphi\left(\frac{s - \mu_j}{\sigma_j}\right) \varphi\left(\frac{s + \delta - \mu_i}{\sigma_i}\right) d\delta ds \quad (5)$$

In this section, we will show that (5) equals

$$\Phi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right). \quad (6)$$

To prove this equality, we will calculate the Fourier transform for (5) and (6) and show that they are the same.<sup>6</sup>

Recall that the Fourier transform for the Normal Cumulative Density function (CDF) is

$$k(\xi) := \int \Phi\left(\frac{x_0 - \mu}{\sigma}\right) e^{-2\pi i \xi x_0} dx_0 = \frac{e^{-2\pi i \xi \mu} e^{-2(\pi \xi \sigma)^2}}{2\pi i \xi} + \frac{1}{2} \delta(\xi) \quad (7)$$

where  $\delta$  is the Dirac Delta function. In Appendix 2, we show how we can derive (7). From this equation, we can use the Inverse Fourier Transform to express the Normal CDF as

$$\Phi\left(\frac{x_0 - \mu}{\sigma}\right) = \int \frac{e^{2\pi i \xi (x_0 - \mu)} e^{-2(\pi \xi \sigma)^2}}{2\pi i \xi} d\xi + \frac{1}{2}. \quad (8)$$

From (7) we see that the Inverse Fourier Transform for (5) equals

$$\Phi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right) = \int \frac{e^{2\pi i \xi (\mu_i - \mu_j)} e^{-2(\pi \xi \sqrt{\sigma_i^2 + \sigma_j^2})^2}}{2\pi i \xi} d\xi + \frac{1}{2}. \quad (9)$$

We will show that the Fourier transform for (5) has the same form as (9). Let  $k(\xi)$  represent the Fourier transform for  $\varphi\left(\frac{s - \mu_j}{\sigma_j}\right)$  and replace this term in (3) with its Fourier transform:

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<sup>6</sup>Whenever two functions have identical Fourier transforms, we know that they are equal.



$$\begin{aligned}
p_{i,j} &= \int \int_0^\infty \varphi\left(\frac{s - \mu_j}{\sigma_j}\right) \varphi\left(\frac{s + \delta - \mu_i}{\sigma_i}\right) d\delta ds \\
&= \int \int_0^\infty \left( \int k(\xi) e^{2\pi i \xi s} d\xi \right) \varphi\left(\frac{s + \delta - \mu_i}{\sigma_i}\right) d\delta ds \\
&\quad (\text{substitute the Inverse Fourier transform}) \\
&= \int k(\xi) \int_0^\infty \int e^{2\pi i \xi s} \varphi\left(\frac{s + \delta - \mu_i}{\sigma_i}\right) ds d\delta d\xi \\
&= \int k(\xi) \int_0^\infty e^{2\pi i \xi (\mu_i - \delta)} e^{-2(\pi \xi \sigma_i)^2} d\delta d\xi \\
&\quad (\text{similar to the derivation for the Fourier transform of the Normal PDF}) \\
&= \int k(\xi) e^{2\pi i \xi \mu_i} e^{-2(\pi \xi \sigma_i)^2} \int_0^\infty e^{-2\pi i \xi \delta} d\delta d\xi \\
&= \int \frac{k(\xi) e^{2\pi i \xi \mu_i} e^{-2(\pi \xi \sigma_i)^2}}{2\pi i \xi} d\xi \\
&= \int \frac{e^{-2\pi i \xi \mu_j} e^{-2(\pi \xi \sigma_j)^2} e^{2\pi i \xi \mu_i} e^{-2(\pi \xi \sigma_i)^2}}{2\pi i \xi} d\xi \\
&\quad (\text{expand } k(\xi)) \\
&= \int \frac{e^{2\pi i \xi (\mu_i - \mu_j)} e^{-2(\pi \xi \sqrt{\sigma_i^2 + \sigma_j^2})^2}}{2\pi i \xi} d\xi.
\end{aligned}$$

All that remains is to set the constant of integration, which in this instance is  $\frac{1}{2}$ . Thus, we see that the Fourier inverse transform for (5) is equivalent to the inverse transform for (6).

## Appendix 2: The Fourier Transforms of the Normal PDF and CDF

In this section, we will show how we can calculate the Fourier Transform for the Normal Probability Density Function (PDF) and Cumulative Density Function (CDF).

We can calculate the Fourier Transform for the PDF by expanding terms and completing the square as follows

$$\begin{aligned}
k(\xi) &:= \int \varphi\left(\frac{x-\mu}{\sigma}\right) e^{-2\pi i \xi x} dx \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{-2\pi i \xi x} dx \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int e^{-\frac{1}{2\sigma^2}(x^2-2\mu x+\mu^2)-2\pi i \xi x} dx \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int e^{-\frac{1}{2\sigma^2}(x^2-2\mu x+\mu^2+4\pi i \xi \sigma^2 x)} dx \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int e^{-\frac{1}{2\sigma^2}(x^2-2x(\mu-2\pi i \xi \sigma^2)+(\mu-2\pi i \xi \sigma^2)^2-(\mu-2\pi i \xi \sigma^2)^2+\mu^2)} dx \\
&\quad \text{(complete the square)} \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int e^{-\frac{1}{2\sigma^2}((x-(\mu-2\pi i \xi \sigma^2))^2-(\mu-2\pi i \xi \sigma^2)^2+\mu^2)} dx \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int e^{-\frac{1}{2\sigma^2}((x-(\mu-2\pi i \xi \sigma^2))^2+4\mu\pi i \xi \sigma^2+4(\pi \xi \sigma^2)^2)} dx \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int e^{-\frac{1}{2\sigma^2}((x-(\mu-2\pi i \xi \sigma^2))^2-2\mu\pi i \xi-2(\pi \xi \sigma^2)^2)} dx \\
&= e^{-2\mu\pi i \xi-2(\pi \xi \sigma^2)^2} \int \varphi\left(\frac{x-(\mu-2\pi i \xi \sigma^2)}{\sigma}\right) dx \\
&= e^{-2\mu\pi i \xi-2(\pi \xi \sigma^2)^2}.
\end{aligned}$$

We can then use this equation to calculate the Inverse Fourier transform for the Normal PDF.

$$\begin{aligned}
\varphi\left(\frac{x-\mu}{\sigma}\right) &= \int k(\xi) e^{2\pi i \xi x} d\xi \\
&= \int e^{-2\mu\pi i \xi-2(\pi \xi \sigma^2)^2} e^{2\pi i \xi x} d\xi \\
&= \int e^{2\pi i \xi(x-\mu)-2(\pi \xi \sigma^2)^2} d\xi
\end{aligned}$$

We can actually simplify this expression further. To do so, note that the integral is quasi symmetric about  $\xi = 0$ . Rewrite the integral so that it spans from 0 to  $\infty$  and move the expression for both  $+\xi$  and  $-\xi$  into it. Next, expand the complex exponential into sines and cosines. The sine terms will cancel out. The result will be

$$\varphi\left(\frac{x-\mu}{\sigma}\right) = 2 \int_0^\infty \cos(2\pi \xi(x-\mu)) e^{-2(\pi \xi \sigma^2)^2} d\xi$$

From the above calculations, we see that the Inverse Fourier transform for (3) must equal

$$\varphi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}\right) = \int e^{2\pi i \xi (\mu_i - \mu_j)} e^{-2(\pi \xi \sqrt{\sigma_i^2 + \sigma_j^2})^2} d\xi. \quad (10)$$

We can use the Integration Property of the Fourier Transform to calculate the Fourier Transform for the Normal CDF. Given a continuous function  $f(x)$ , the Integration Property says

$$\mathcal{F}\left(\int_{-\infty}^{x_0} f(x) dx\right) = \frac{\mathcal{F}(f(x))}{2\pi i \xi} + c \delta(\xi) \quad (11)$$

Where  $\mathcal{F}$  denotes the Fourier Transform operator,  $\delta$  represents the Dirac delta function, and  $c$  is a constant that depends on  $f$ . In the standard proof the integral in (11) is represented by a convolution of  $f$  and the step-function. We then use the Convolution Property of the Fourier Transform to express the Fourier Transform as the product of the Fourier Transform of  $f$  and the step function. The Fourier Transform of the step function reduces to a sum involving the Dirac Delta function. A detailed derivation can be found in <sup>7</sup>.

Applying the Integration Property of the Fourier Transform to the Normal CDF gives

$$\Phi\left(\frac{x_0 - \mu}{\sigma}\right) = \int \frac{e^{2\pi i \xi (x_0 - \mu)} e^{-2(\pi \xi \sigma)^2}}{2\pi i \xi} d\xi + \frac{1}{2}. \quad (12)$$

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<sup>7</sup>Bevelacqua, P. (n.d.). *The Integration Property of the Fourier Transform*. The Fourier Transform.com. <https://www.thefouriertransform.com/transform/integration.php>